

Activations: ReLU, GELU, SwiGLU

The bend inside every MLP · video 2:45

This lesson answers one question: what is the “bend” in the middle of every MLP, and why have models moved from ReLU to GELU to **SwiGLU**? Without an activation, the MLP's two matrices collapse into one. GPT-2's GELU keeps small negative values that the model relies on (swap in ReLU and the loss jumps from 4.13 to 7.23). SwiGLU adds a learned gate; in a tiny GPT with about the same parameter budget it beat GELU, which beat ReLU, in all three seeds.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 6).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. Why an MLP needs a bend	▶ 0:08–0:46	2	☐
2. GPT-2 relies on GELU's dip	▶ 0:46–1:09	2	☐
3. SwiGLU: a gated MLP	▶ 1:09–1:42	2	☐
4. Results, dead neurons, and the code	▶ 1:42–2:32	2	☐

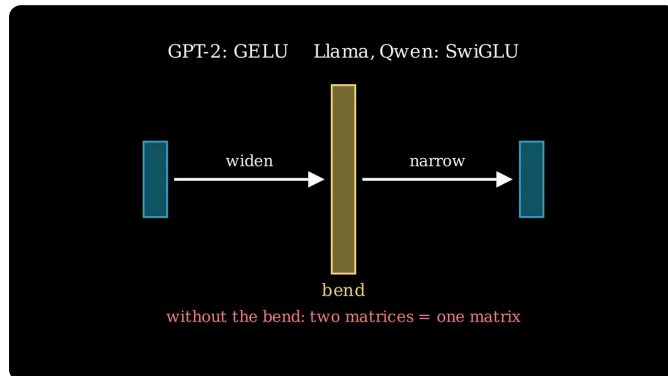
Before you start: How LLMs Work episode 8 (the MLP) and Deep Dive episode 13 (the residual stream). Code: `code/d15_activations` (GPT-2 small and Qwen2.5-0.5B, about 1.5 GB, plus nine tiny models, about 40 minutes on a CPU).

CONCEPT 1 OF 4

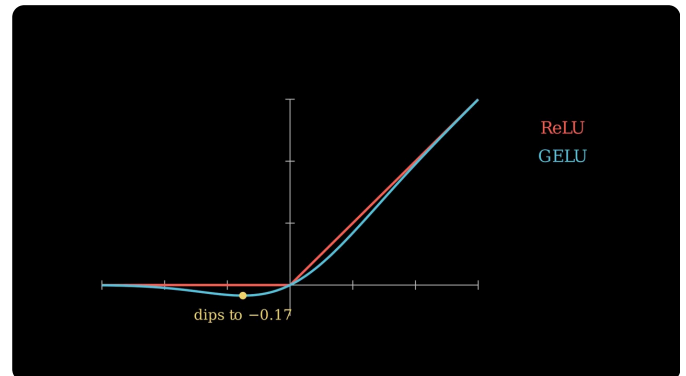
Why an MLP needs a bend

▶ Video 0:08–0:46

Every MLP has the same shape: **widen** (768 → 3,072 in GPT-2), **bend**, **narrow** (back to 768). The bend is the activation function. Without it, $W_2(W_1x) = (W_2W_1)x$: the two matrices would collapse into one, and the MLP could only compute linear functions.



▶ 0:26 Widen, bend, narrow. Without the bend, two matrices collapse into one.



▶ 0:44 ReLU (zero for negatives) and GELU (smooth, dipping to -0.17).

ReLU is the simplest bend: negative numbers become 0, positive ones pass. **GELU** is a smooth version: large positive numbers pass, large negative ones fade to 0, but in between it dips below zero, down to -0.17 (near $x = -0.75$). At $x = -1$: ReLU 0, GELU -0.159 , SiLU -0.269 .

KEY IDEA

The activation is the only nonlinear step in the MLP; its exact shape decides what passes and what is suppressed.

Check 1: Why an MLP needs a bend

Answer all 2 questions before looking at the answers. Pass mark: all correct.

1.1 CALCULATION

What are $\text{ReLU}(0.5)$, $\text{GELU}(0.5)$ and $\text{SiLU}(0.5)$?

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1.2 SHORT ANSWER

Why can't an MLP without an activation compute anything a single matrix can't?

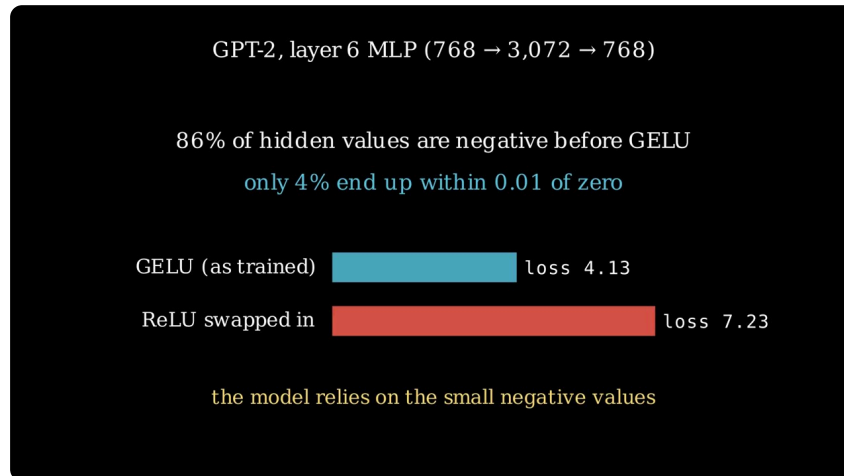
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CONCEPT 2 OF 4

GPT-2 relies on GELU's dip

▶ Video 0:46–1:09

In GPT-2's layer 6 MLP, **86%** of the 3,072 hidden values are negative before GELU. Most of them do not become zero: only **4%** end up within 0.01 of zero. Swap every GELU for ReLU, without retraining, and the loss on 512 tokens of Shakespeare jumps from **4.13** to **7.23**. The model relies on those small negative values.



▶ 1:07 86% of the hidden values are negative before GELU; swapping in ReLU raises the loss from 4.13 to 7.23.

KEY IDEA

A trained model is tuned to its exact activation; the negative side of GELU carries information.

Check 2: GPT-2 relies on GELU's dip

Answer all 2 questions before looking at the answers. Pass mark: all correct.

2.1 TRUE OR FALSE

“Because most hidden values are negative, GELU outputs are mostly exactly zero.”

True ☐ False ☐

2.2 MULTIPLE CHOICE

Swapping GPT-2's GELU for ReLU without retraining...

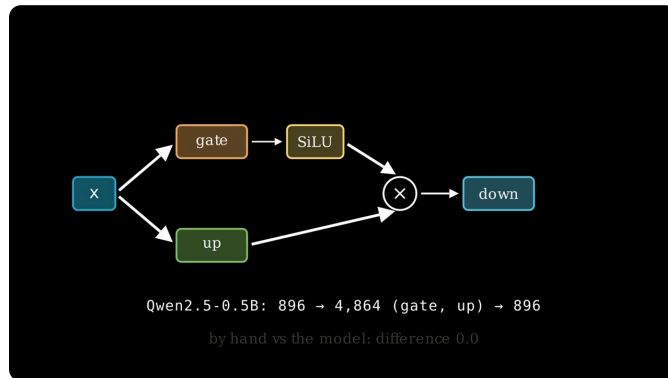
- A. leaves the loss unchanged
- B. slightly improves it
- C. raises it from 4.13 to 7.23
- D. makes it NaN

CONCEPT 3 OF 4

SwiGLU: a gated MLP

▶ Video 1:09-1:42

SwiGLU adds a gate. Two matrices go up: a **gate** and an **up** projection. The gate passes through **SiLU** ($x \cdot \text{sigmoid}(x)$, a smooth bend like GELU) and multiplies the up projection number by number; a third matrix goes **down**. In Qwen2.5-0.5B that is $896 \rightarrow 4,864 \rightarrow 896$, with 13,074,432 parameters per MLP; the hand computation matches the model exactly (difference 0.0).



▶ 1:31 SwiGLU: $\text{down}(\text{SiLU}(\text{gate}(x)) \times \text{up}(x))$;
Qwen2.5-0.5B: $896 \rightarrow 4,864 \rightarrow 896$, matched exactly.

GELU / ReLU MLP:	$128 \rightarrow 512 \rightarrow 128$	2 matrices	131,072 parameters
SwiGLU MLP:	$128 \rightarrow 344 (\times 2) \rightarrow 128$	3 matrices	132,096 parameters
narrower middle (2/3 of the width), about the same parameters			

▶ 1:41 Three matrices with a narrower middle (344 instead of 512) keep the parameter count about the same.

Three matrices instead of two would be an unfair comparison, so SwiGLU uses a narrower middle, about two thirds of the width: in the tiny GPT, 344 instead of 512 (132,096 parameters vs 131,072).

KEY IDEA

SwiGLU = $\text{SiLU}(\text{gate} \cdot x) \odot (\text{up} \cdot x)$, then down: each hidden unit is a product of two learned signals.

Check 3: SwiGLU: a gated MLP

Answer all 2 questions before looking at the answers. Pass mark: all correct.

3.1 CALCULATION

Qwen2.5-0.5B's MLP has three $896 \times 4,864$ matrices and no biases. How many parameters?

3.2 CALCULATION

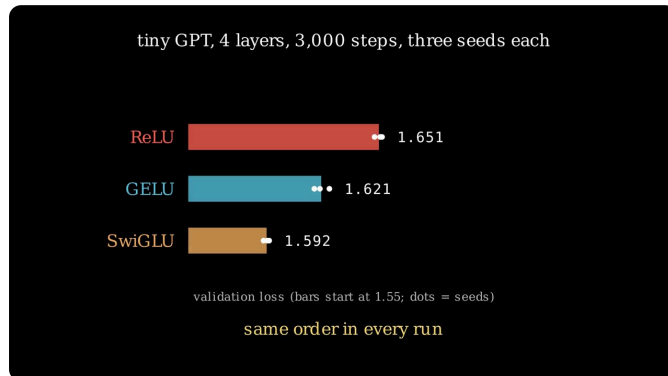
With $D = 128$, a GELU MLP ($128 \rightarrow 512 \rightarrow 128$, no biases) has 131,072 parameters. What middle width h gives a SwiGLU MLP (3 matrices of $128 \times h$) the same count?

CONCEPT 4 OF 4

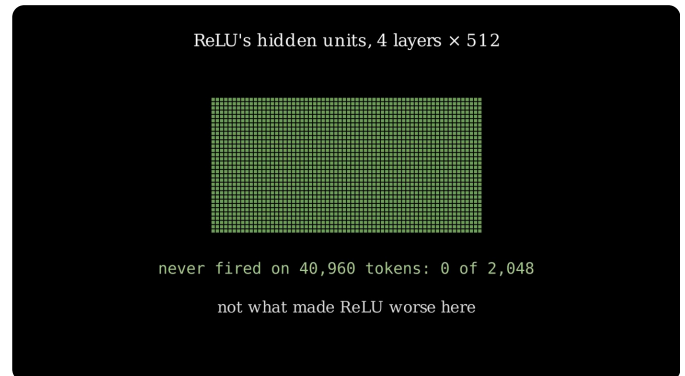
Results, dead neurons, and the code

▶ Video 1:42-2:32

The same 4-layer tiny GPT, trained 3,000 steps with each activation, three seeds each. Validation loss: ReLU 1.653 / 1.649 / 1.652 (mean **1.651**), GELU 1.620 / 1.617 / 1.625 (**1.620**), SwiGLU 1.590 / 1.592 / 1.593 (**1.591**). The same order in every run: a small gain, at no extra cost.



▶ 1:58 Three seeds each: ReLU 1.651, GELU 1.620, SwiGLU 1.591 (means); the same order in every run.



▶ 2:11 In the ReLU model, all 2,048 hidden units fired at least once on 40,960 tokens.

A classic worry with ReLU is **dead neurons**, units that never fire again. In the trained ReLU model, all 2,048 hidden units fired at least once on 40,960 validation tokens, so at this scale that is not what made ReLU worse. One common explanation for the gates' advantage: each hidden unit is a product of two learned signals, so the MLP can switch features on and off depending on the input.

KEY IDEA

Smooth activations beat ReLU, and gating beats both, by a small but consistent margin.

Check 4: Results, dead neurons, and the code

Answer all 2 questions before looking at the answers. Pass mark: all correct.

4.1 PUT IN ORDER

Order the activations from best to worst validation loss in the episode: *GELU · ReLU · SwiGLU*.

4.2 HANDS-ON CODE

Try it yourself. Write SwiGLU's forward pass with plain tensors and check it against Qwen's MLP, as the code does.

```
gate, up = mlp.gate_proj(x), mlp.up_proj(x)
mine = mlp.down_proj(F.silu(gate) * up)
print((mine - mlp(x)).abs().max())
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. Why an MLP needs a bend · rewatch ▶ 0:08-0:46

1.1 0.5, 0.346, 0.311. From the code's table; the smooth functions pass less than ReLU near zero.

1.2 $W_2(W_1x) = (W_2W_1)x$: the product of two matrices is one matrix. Stacking linear maps gives a linear map.

2. GPT-2 relies on GELU's dip · rewatch ▶ 0:46-1:09

2.1 False. Only 4% end up within 0.01 of zero; GELU maps moderate negative values to small negative outputs (down to -0.17).

2.2 C. Measured on 512 tokens of Shakespeare.

3. SwiGLU: a gated MLP · rewatch ▶ 1:09-1:42

3.1 13,074,432. $3 \times 896 \times 4,864$.

3.2 About 341 (the code uses 344). $3 \times 128 \times h = 131,072 \rightarrow h \approx 341.3$; 344 is a round multiple of 8.

4. Results, dead neurons, and the code · rewatch ▶ 1:42-2:32

4.1 SwiGLU (1.591) → GELU (1.620) → ReLU (1.651). Means over three seeds; the same order in each seed.

4.2 0.0. Qwen2's MLP computes exactly $\text{down}(\text{silu}(\text{gate}(x)) * \text{up}(x))$.